pg 2

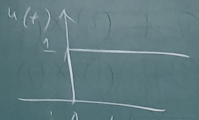
$$F(s) = \int_0^{\infty} e^{-st} f(t)\,dt, \quad s: \text{complex}$$

one-sided / unilateral

Two-sided $\int_{-\infty}^{\infty} e^{-st} f(t)\,dt$ — Transient analysis

$v(t)$, $i(t)$

$v(t) \cdot u(t)$ ← unit-step fn



pg 1

Recap

Capacitor & Inductor "almost linear"

↳ in t-domain either V-I or I-V is linear



In a circuit w/ res + cap + ind (⇐ a circuit w/ only resistor in s-domain!

$\text{Cap} \Leftarrow \frac{1}{sC}\left(\text{Res}\right) + \frac{v(0)}{s}$

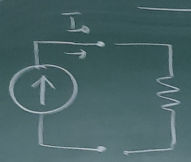
$\text{Ind} \Leftarrow sL\left(\text{Ind}\right) + LI(0)$

↳ in s-domain (Laplace)

⇐ Resistor (s-dependent) + Voltage source

PS2

Ideal current source

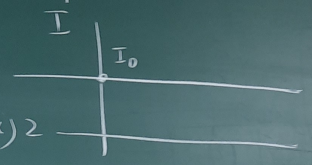


Provides a fixed current (I_0) irrespective of the voltage across it.

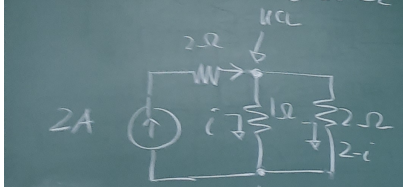
$1\Omega \rightarrow I_0 \cdot 1$

$0.1\Omega \rightarrow I \times 0.1$

0.001Ω



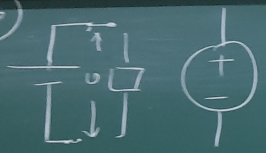
"Infinite resistance"

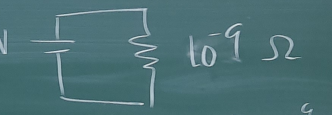


$i \cdot 1 = (2-i)2$

NOT useful write KVL w/ current source in the loop

PS3





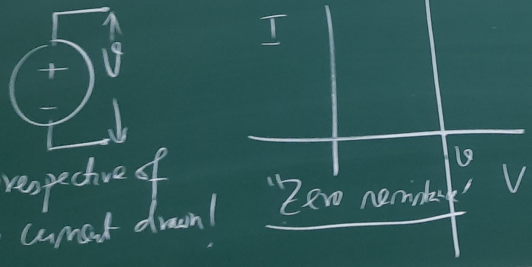
3×10^{-9} A

$V \cdot i = 9 \times 10^{-9}$ W

Battery: 3V, 0.1 mA

ideal voltage source

Ideal voltage source



irrespective of the current drawn!

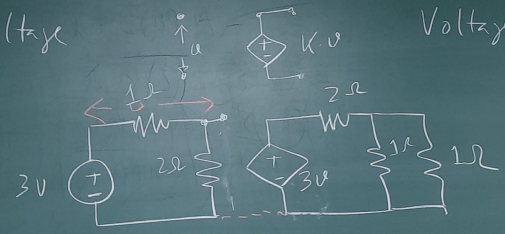
"Zero resistance"

Controlled source (ideal)

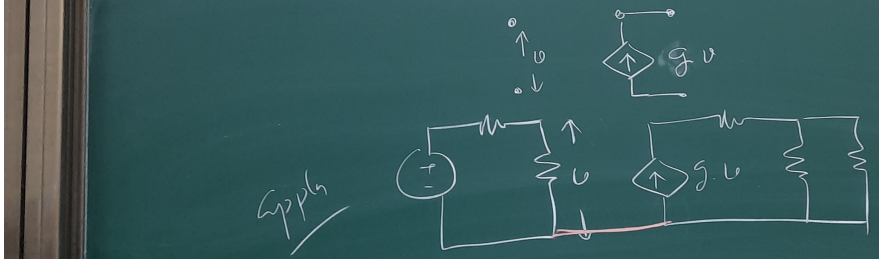
2-terminal (Control) + 2 terminal (source)

Voltage Voltage

Voltage controlled voltage source → VCVS



VCCS (voltage controlled current source)



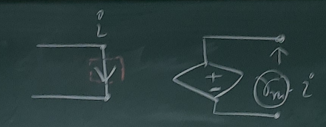
Supply

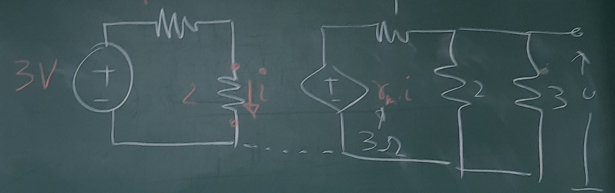
MOSFET

p32

CCVS

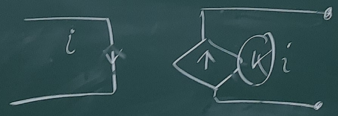
Current controlled voltage source





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Current Controlled current source



Inductor

$$V(t) = L \frac{dI(t)}{dt}$$